

Bank Loan Analysis

1. Project Overview

This project analyzes a bank's consumer loan portfolio using a dataset of 38,576 loan applications. SQL was used to clean the data and answer key business questions, and the results were visualized in an interactive three-page Power BI dashboard. The goal is to give bank stakeholders a clear, data-driven view of loan performance, risk (good vs. bad loans), and the borrower characteristics that drive lending volume.

2. Dataset Summary

- Rows: 38,576
- Columns: 24
- Key Features:
 - Borrower/loan identifiers (id, member_id, application_type)
 - Borrower profile (address_state, emp_title, emp_length, home_ownership, annual_income, verification_status, dti)
 - Loan details (loan_amount, term, int_rate, installment, grade, sub_grade, purpose, total_acc)
 - Loan timeline & status (issue_date, last_payment_date, next_payment_date, last_credit_pull_date, loan_status)
 - Repayment outcome (total_payment)

3. Data Analysis using SQL

Performed structured analysis in PostgreSQL to calculate the core KPIs and breakdowns later surfaced on the Power BI dashboard.

3.1 Overall KPIs

KPI	Total	MTD (Dec)	PMTD (Nov)	MoM Change
Loan Applications	38,576	4,314	4,035	▲ 6.9%
Total Funded Amount	\$435.7M	\$53.9M	\$47.7M	▲ 13.0%
Total Amount Received	\$473.0M	\$58.0M	\$50.1M	▲ 15.8%
Avg. Interest Rate	12.05%	12.36%	11.94%	▲ 3.5%
Avg. DTI	13.33%	13.67%	13.30%	▲ 2.8%

3.2 Good Loan vs. Bad Loan

Loans with status Fully Paid or Current were classified as Good Loans; loans Charged Off were classified as Bad Loans.

Category	Applications	% of Total	Funded Amount	Amount Received
Good Loans	33,243	86.18%	\$370.2M	\$435.7M
Bad Loans	5,333	13.82%	\$65.5M	\$37.2M

Key Insight: the bank funded \$65.5M in loans that were ultimately charged off but recovered only \$37.2M of it — a loss of roughly \$28M on bad loans.

3.3 Loan Status Grid View

Loan Status	Applications	Funded Amount	Amount Received	Avg Rate	Interest	Avg DTI
Fully Paid	32,145	\$351.4M	\$411.6M	11.64%		13.17%
Charged Off	5,333	\$65.5M	\$37.3M	13.88%		14.00%
Current	1,098	\$18.9M	\$24.2M	15.10%		14.72%

- **Risk pricing:** Charged Off loans carry a higher average interest rate (13.88%) — riskier borrowers are priced higher.
- **Active loans:** Current loans show the highest average rate (15.10%), reflecting ongoing active payments.
- **Profitability:** Fully Paid loans returned about \$60M more than was funded, reflecting healthy interest income.

3.4 Monthly Trend

	month_number numeric	month_name text	total_applications bigint	total_funded_amount bigint	total_received_amount numeric
1	1	Jan	2332	25031650	27578836
2	2	Feb	2279	24647825	27717745
3	3	Mar	2627	28875700	32264400
4	4	Apr	2755	29800800	32495533
5	5	May	2911	31738350	33750523
6	6	Jun	3184	34161475	36164533
7	7	Jul	3366	35813900	38827220
8	8	Aug	3441	38149600	42682218
9	9	Sep	3536	40907725	43983948
10	10	Oct	3796	44893800	49399567
11	11	Nov	4035	47754825	50132030
12	12	Dec	4314	53981425	58074380

Applications grew steadily across the year, from 2,332 in January to 4,314 in December — almost doubling over twelve months, alongside parallel growth in funded and received amounts.

3.5 Regional Analysis (Top States)

State	Applications	Funded Amount	Amount Received
CA	6,894	\$78.5M	\$83.9M
NY	3,701	\$42.1M	\$46.1M
TX	2,664	\$31.2M	\$34.4M
FL	2,773	\$30.0M	\$31.6M
NJ	1,822	\$21.7M	\$23.4M

California dominates lending volume with 6,894 applications and \$78.5M funded, nearly double the next closest state (New York).

3.6 Loan Term, Employment Length & Purpose

- By term:

	term character varying (20)	total_applications bigint	total_funded_amount bigint	total_received_amount numeric
1	36 months	28237	273041225	294709458
2	60 months	10339	162715850	178361475

36-month terms are far more common (28,237 applications) than 60-month terms (10,339).

- By employment length:

	emp_length character varying (20)	total_applications bigint	total_funded_amount bigint	total_received_amount numeric
1	10+ years	8870	116115950	125871616
2	< 1 year	4575	44210625	47545011
3	2 years	4382	44967975	49206961
4	3 years	4088	43937850	47551832
5	4 years	3428	37600375	40964850
6	5 years	3273	36973625	40397571
7	1 year	3229	32883125	35498348
8	6 years	2228	25612650	27908658
9	7 years	1772	20811725	22584136
10	8 years	1476	17558950	19025777
11	9 years	1255	15084225	16516173

Borrowers with 10+ years of employment account for the most loans (8,870) — a sign of more stable, established applicants.

- **By purpose:**

	purpose character varying (50)	total_applications bigint	total_funded_amount bigint	total_received_amount numeric
1	Debt consolidation	18214	232459675	253801871
2	credit card	4998	58885175	65214084
3	other	3824	31155750	33289676
4	home improvement	2876	33350775	36380930
5	major purchase	2110	17251600	18676927
6	small business	1776	24123100	23814817
7	car	1497	10223575	11324914
8	wedding	928	9225800	10266856
9	medical	667	5533225	5851372
10	moving	559	3748125	3999899
11	house	366	4824925	5185538
12	vacation	352	1967950	2116738
13	educational	315	2161650	2248380
14	renewable_energy	94	845750	898931

Debt consolidation is by far the leading purpose (18,214 applications, \$232.5M funded), followed by credit card refinancing (4,998 applications).

- **By home ownership:**

	home_ownership character varying (20)	total_applications bigint	total_funded_amount bigint	total_received_amount numeric
1	RENT	18439	185768475	201823056
2	MORTGAGE	17198	219329150	238474438
3	OWN	2838	29597675	31729129
4	OTHER	98	1044975	1025257
5	NONE	3	16800	19053

Renters (18,439) slightly outnumber mortgage holders (17,198) among applicants.

4. Power BI Dashboard

A three-page interactive Power BI report was built on top of the SQL Server data, with slicers for Purpose, Grade, and State that drive every page.

4.1 Summary Page

Headline KPIs (applications, funded amount, amount received, average interest rate, average DTI) alongside a Good Loan vs. Bad Loan breakdown and the Loan Status grid.

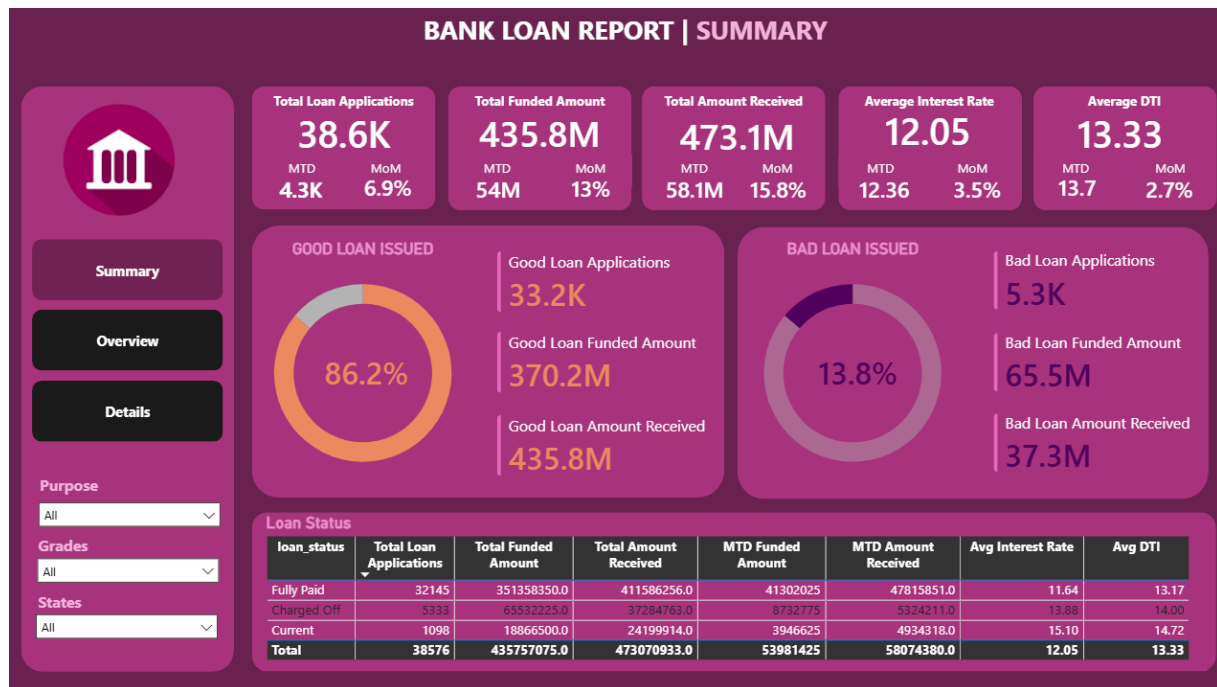
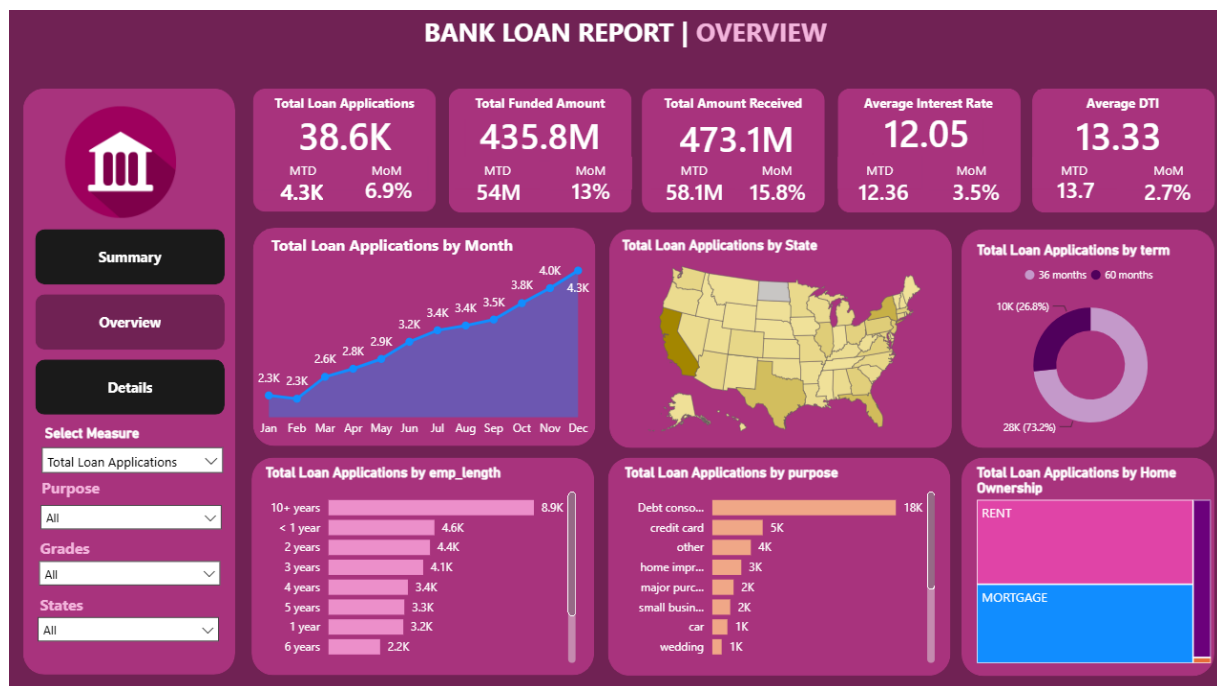


Figure 1 — Bank Loan Report: Summary page

4.2 Overview Page

Visual breakdown of lending activity by month, U.S. state, term, employment length, purpose, and home ownership.



4.3 Details Page

A row-level grid giving analysts a one-stop view of individual loan records (purpose, home ownership, grade, issue date, funded amount, interest rate, installment, and amount collected).

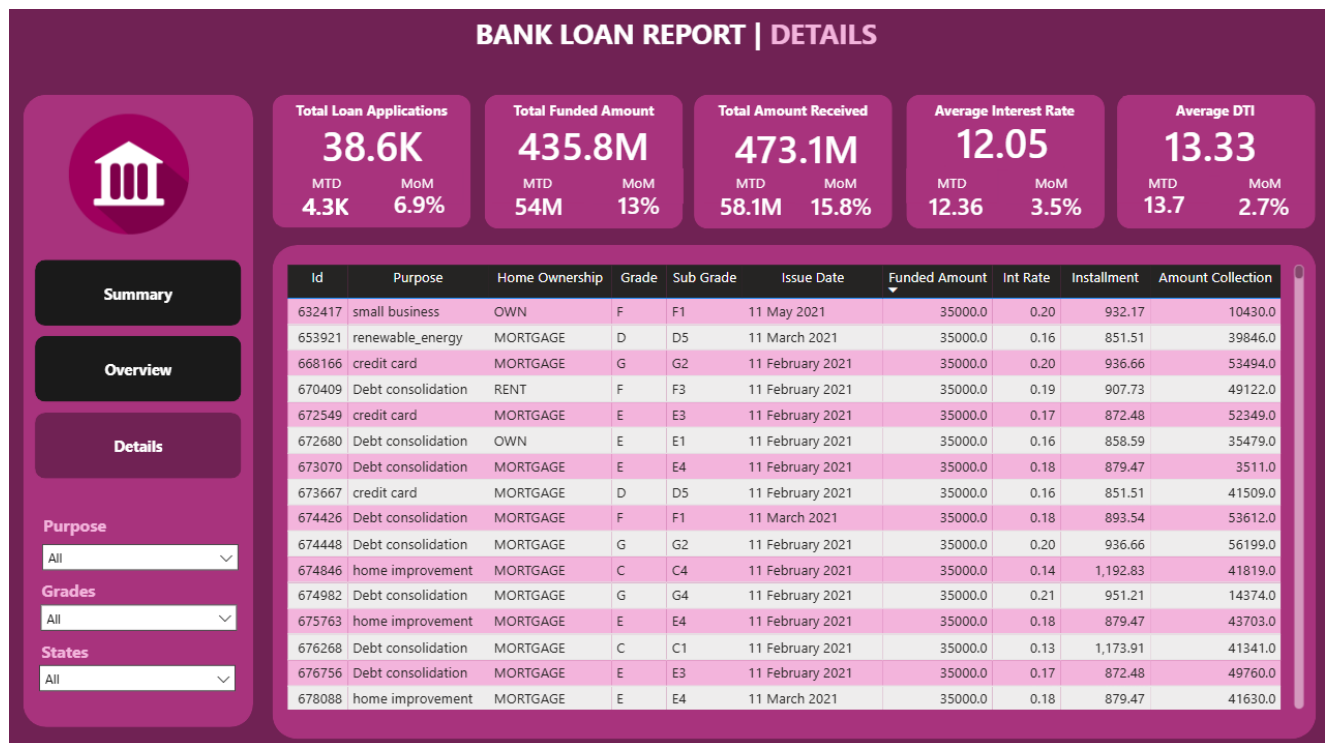


Figure 3 — Bank Loan Report: Details page

5. Business Recommendations

- **Reduce bad-loan exposure** – With 13.82% of applications charged off and a ~\$28M shortfall on those loans, tighten underwriting criteria (DTI thresholds, grade-based limits) for the riskiest borrower segments.
- **Reward stable borrowers** – Loans with 10+ years of employment history perform reliably; consider preferential rates or faster approval for this segment.
- **Double down on top purposes** – Debt consolidation and credit card refinancing dominate demand; tailor marketing and product bundling around these two purposes.
- **Focus regional efforts** – California, New York, Texas, and Florida drive the bulk of volume; prioritize regional marketing spend and local partnerships there.
- **Plan for continued growth** – Loan volume nearly doubled year-over-year; ensure underwriting and servicing capacity scale ahead of continued growth.
- **Watch active loans** – Current loans carry the highest average interest rate (15.10%); monitor this cohort closely as it migrates to Fully Paid or Charged Off.